Annual Report - General Order No. R5-2007-0035
Reporting period 01/01/2023 to 12/31/2023.

DAIRY FACILITY INFORMATION

A. NAME OF DAIRY OR BUSINESS OPERATING THE DAIRY: G. Simoes Dairy, Inc.

Physical address of dairy:

11758 Carroll RD	Elk Grove	Sacramento	95757
Number and Street	City	County	Zip Code

Street and nearest cross street (if no address):

Date facility was originally placed in operation: 01/01/1976

Regional Water Quality Control Board Basin Plan designation: Sacramento River Basin

County Assessor Parcel Number(s) for dairy facility:

X132-X240-X051-XXXX

B. OPERATORS

George, Simoes

Operator name: George, Simoes

Telephone no.: (916) 826-2495 (Landline) (Cellular)

9792 Diablo Vista AVE	Galt	CA	95632
Mailing Address Number and Street	City	State	Zip Code

This operator is responsible for paying permit fees.

C. OWNERS

George, Simoes

Legal owner name: George, Simoes

Telephone no.: (916) 826-2495 (Landline) (Cellular)

9792 Diablo Vista AVE	Galt	CA	95632
Mailing Address Number and Street	City	State	Zip Code

This owner is responsible for paying permit fees.

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AVAILABLE NUTRIENTS

A. HERD INFORMATION

	Milk Cows	Dry Cows	Bred Heifers (15-24 mo.)	Heifers (7-14 mo. to breeding)	Calves (4-6 mo.)	Calves (0-3 mo.)
Number open confinement	0	0	350	600	0	0
Number under roof	0	0	0	0	0	0
Maximum number	0	0	350	600	0	0
Average number	0	0	350	600	0	0
Avg live weight (lbs)	0	0	1,000	750		

Predominant milk cow breed: Holstein

Average milk production: 1 *pounds per cow per day*

B. MANURE GENERATED

Total manure excreted by the herd: 9,438.13 *tons per reporting period*

Total nitrogen from manure: 90,155.00 *lbs per reporting period*

After ammonia losses (30% loss applied): 63,108.50 *lbs per reporting period*

Total phosphorus from manure: 15,257.00 *lbs per reporting period*

Total potassium from manure: 1.00 *lbs per reporting period*

Total salt from manure: 0.00 *lbs per reporting period*

C. PROCESS WASTEWATER GENERATED

Process wastewater generated: 1,770,000 *gallons*

Total nitrogen generated: 1,968.78 *lbs*

Total phosphorus generated: 964.52 *lbs*

Total potassium generated: 8,714.68 *lbs*

Total salt generated: 40,471.58 *lbs*

	1,770,000	*gallons applied*
+	0	*gallons exported*
-	0	*gallons imported*
=	1,770,000	*gallons generated*

D. FRESH WATER SOURCES

Source Description	Type
Ag Well #1	Ground water
Ag Well #2	Ground water
DW #1	Ground water

E. SUBSURFACE (TILE) DRAINAGE SOURCES

No subsurface (tile) drainage sources entered.

F. NUTRIENT IMPORTS

No dry manure nutrient imports entered.

No process wastewater nutrient imports entered.

No commercial or other nutrient imports entered.

G. NUTRIENT EXPORTS

No solid nutrient exports entered.

No liquid nutrient exports entered.

APPLICATION AREA

A. LIST OF LAND APPLICATION AREAS

Field name	Controlled acres	Cropable acres	Total harvests	Type of waste applied	Parcel number
Field 1	80	80	0	none	X132-X240-X031-XXXX
Field 2	60	60	0	none	X132-0240-X051-XXXX
Field 3	60	60	2	manure	X132-0240-X051-XXXX
Field 4	60	60	0	none	X132-0240-X051-XXXX
Field 5	35	35	1	both	X132-0240-X051-XXXX
Totals for areas that were used for application	95	95	3		
Totals for areas that were not used for application	200	200	0		
Land application area totals	295	295	3		

B. CROPS AND HARVESTS

Field 3

Field name: Field 3

10/27/2022: Oats, silage-soft dough

Crop: Oats, silage-soft dough Acres planted: 60 Plant date: 10/27/2022

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
05/02/2023	505.30 *ton*	Dry-weight		75.2	12,000.00	2,380.00	18,400.00		12.90
05/02/2023	505.30 *ton*	Dry-weight		73.8	11,600.00	2,360.00	18,600.00		20.60

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	16.00	160.00	25.60	132.80	0.00
Total actual harvest content	16.84	101.32	20.36	158.94	1,447.92

Field 5

Field name: Field 5

Field 5

05/15/2023: Corn, silage

Crop: Corn, silage Acres planted: 35 Plant date: 05/15/2023

Harvest date	Yield	Reporting basis	Density (lbs/cu ft)	Moisture (%)	N (mg/kg)	P (mg/kg)	K (mg/kg)	Salt (mg/kg)	TFS (%)
09/18/2023	1,050.00 *ton*	Dry-weight		71.0	11,500.00	3,320.00	16,100.00		6.70

	Yield (tons/acre)	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Salt (lbs/acre)
Anticipated harvest content	27.00	216.00	51.30	248.40	0.00
Total actual harvest content	30.00	200.10	57.77	280.14	1,165.80

NUTRIENT BUDGET

A. LAND APPLICATIONS

Field 3 - 10/27/2022: Oats, silage-soft dough

Field name: Field 3

Crop: Oats, silage-soft dough — Plant date: 10/27/2022

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
10/24/2022	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	116.73	45.38	328.23	2,968.55	360.00 *ton*
Application event totals		116.73	45.38	328.23	2,968.55	

Field 5 - 05/15/2023: Corn, silage

Field name: Field 5

Crop: Corn, silage — Plant date: 05/15/2023

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
05/11/2023	Broadcast/incorporate	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Manure Application	Corral solids	134.15	40.24	183.64	0.00	280.00 *ton*
Application event totals		134.15	40.24	183.64	0.00	

05/16/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #2	Ground water	1.28	0.00	0.00	0.00	4,380,000.00 *gal*
Application event totals		1.28	0.00	0.00	0.00	

05/31/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #2	Ground water	1.16	0.00	0.00	0.00	3,960,000.00 *gal*
Application event totals		1.16	0.00	0.00	0.00	

Field 5 - 05/15/2023: Corn, silage

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	25.95	18.06	149.61	586.61	885,000.00 *gal*
Ag Well #2	Ground water	0.88	0.00	0.00	0.00	3,000,000.00 *gal*
Application event totals		26.83	18.06	149.61	586.61	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
06/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #2	Ground water	1.06	0.00	0.00	0.00	3,600,000.00 *gal*
Application event totals		1.06	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/15/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #2	Ground water	1.06	0.00	0.00	0.00	3,600,000.00 *gal*
Application event totals		1.06	0.00	0.00	0.00	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
07/30/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Lagoon Application	Process wastewater	30.30	9.50	99.39	569.73	885,000.00 *gal*
Ag Well #2	Ground water	0.88	0.00	0.00	0.00	3,000,000.00 *gal*
Application event totals		31.18	9.50	99.39	569.73	

Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/14/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #2	Ground water	1.06	0.00	0.00	0.00	3,600,000.00 *gal*
Application event totals		1.06	0.00	0.00	0.00	

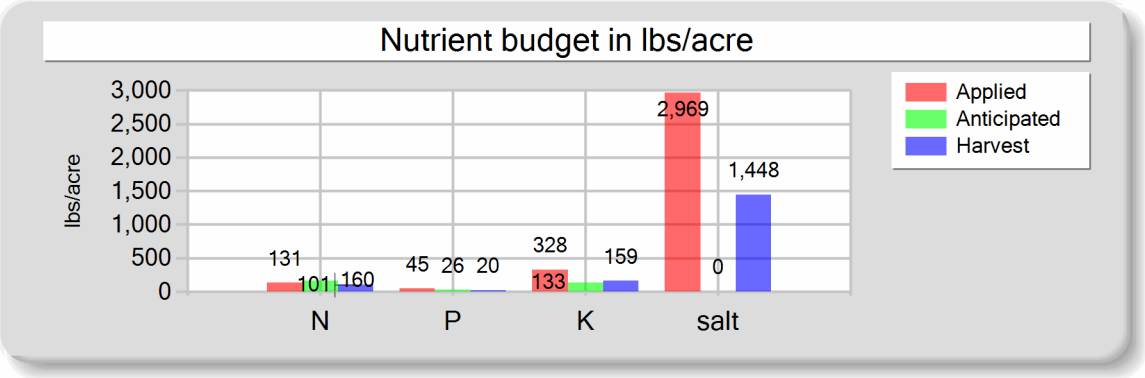
Application date	Application method	Precipitation 24 hours prior	Precipitation during application	Precipitation 24 hours following
08/29/2023	Surface (irrigation)	No precipitation	No precipitation	No precipitation

Source description	Material type	N (lbs/acre)	P (lbs/acre)	K (lbs/acre)	Salt (lbs/acre)	Amount
Ag Well #2	Ground water	1.06	0.00	0.00	0.00	3,600,000.00 *gal*
Application event totals		1.06	0.00	0.00	0.00	

B. NUTRIENT BUDGET

Field 3 - 10/27/2022: Oats, silage-soft dough

Field name: Field 3 Crop: Oats, silage-soft dough Plant date: 10/27/2022



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	116.73	45.38	328.23	2,968.55
Process wastewater	0.00	0.00	0.00	0.00
Fresh water	0.00	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	130.73	45.38	328.23	2,968.55
Anticipated crop nutrient removal	160.00	25.60	132.80	0.00
Actual crop nutrient removal	101.32	20.36	158.94	1,447.92
Nutrient balance	29.42	25.02	169.29	1,520.63
Applied to removed ratio	1.29	2.23	2.07	2.05

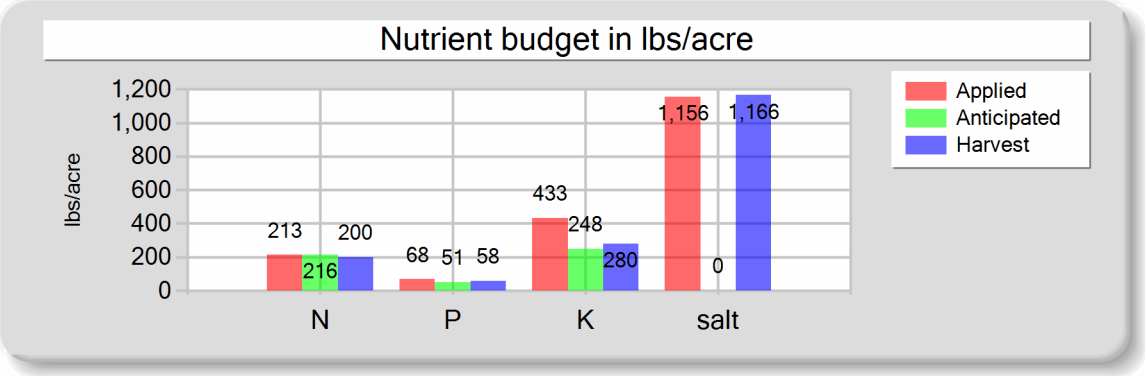
Fresh water applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Process wastewater applied
0.00 *gallons*
0.00 *acre-inches*
0.00 *inches/acre*

Total harvests for the crop
2 *harvests*

Field 5 - 05/15/2023: Corn, silage

Field name: Field 5 Crop: Corn, silage Plant date: 05/15/2023



	Total N (lbs/acre)	Total P (lbs/acre)	Total K (lbs/acre)	Total salt (lbs/acre)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	134.15	40.24	183.64	0.00
Process wastewater	56.25	27.56	248.99	1,156.33
Fresh water	8.43	0.00	0.00	0.00
Atmospheric deposition	14.00	0.00	0.00	0.00
Total nutrients applied	212.83	67.80	432.63	1,156.33
Anticipated crop nutrient removal	216.00	51.30	248.40	0.00
Actual crop nutrient removal	200.10	57.77	280.14	1,165.80
Nutrient balance	12.73	10.03	152.49	-9.47
Applied to removed ratio	1.06	1.17	1.54	0.99

Fresh water applied	
28,740,000.00	*gallons*
1,058.40	*acre-inches*
30.24	*inches/acre*

Process wastewater applied	
1,770,000.00	*gallons*
65.18	*acre-inches*
1.86	*inches/acre*

Total harvests for the crop	
1	*harvests*

NUTRIENT ANALYSES

A. MANURE ANALYSES

T2313021A

Sample and source description: T2313021A

Sample date: 11/09/2022 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 47.7 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	18,600.00	7,230.00	52,300.00	12,100.00	7,980.00	10,000.00	6,200.00	1,260.00		47.30
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

T3130026A

Sample and source description: T3130026A

Sample date: 05/10/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 18.6 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	10,300.00	3,090.00	14,100.00							0.00
DL	1.00	1.00	1.00							1.00

T3265028C

Sample and source description: T3265028C

Sample date: 09/22/2023 Material type: Corral solids Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 48.3 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Calcium (mg/kg)	Magnesium (mg/kg)	Sodium (mg/kg)	Sulfur (mg/kg)	Chloride (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	21,200.00	6,770.00	38,700.00	12,000.00	8,460.00	8,330.00	9,490.00	1,160.00		36.60
DL	1.00	1.00	1.00	1.00	1.00	1.00	1.00	1.00		1.00

B. PROCESS WASTEWATER ANALYSES

T3059011A

Sample and source description: T3059011A

Sample date: 02/28/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	112.00	53.20	0.00	0.00	56.60	467.00								2,580.00	2,300
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3130026B

Sample and source description: T3130026B

Sample date: 05/10/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 0.00

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	123.00	53.20	0.00	0.00	85.60	709.00								3,860.00	2,780
DL	1.00	1.00	1.00	1.00	1.00	1.00								100.00	1

T3265028B

Sample and source description: T3265028B

Sample date: 09/22/2023 Material type: Process wastewater Source of analysis: Lab analysis pH: 7.70

	Kjeldahl-N (mg/L)	NH4-N (mg/L)	NH3-N (mg/L)	Nitrate-N (mg/L)	Total P (mg/L)	Total K (mg/L)	Calcium (mg/L)	Magnes. (mg/L)	Sodium (mg/L)	Bicarb. (mg/L)	Carb. (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	143.00	112.00	0.00	0.58	45.00	471.00	110.00	85.70	126.00	1,460.00	0.00	79.60	337.00	3,920.00	2,700
DL	1.00	1.00	1.00	0.50	1.00	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

C. FRESH WATER ANALYSES

Ag Well #1

Ag Well #1

T3314016B

Sample description: T3314016B

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	2.52	0.00	2.52								616.00	0
DL	1.00	0.50	1.00								100.00	1

Ag Well #2

T3314016C

Sample description: T3314016C

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	1.23	0.00	1.23								510.00	0
DL	1.00	0.50	1.00								100.00	1

DW #1

T3314016A

Sample description: T3314016A

Sample date: 11/10/2023 Source of analysis: Lab analysis

	Total N (mg/L)	NH4-N (mg/L)	Nitrate-N (mg/L)	Calcium (mg/L)	Magnesium (mg/L)	Sodium (mg/L)	Bicarbonate (mg/L)	Carbonate (mg/L)	Sulfate (mg/L)	Chloride (mg/L)	EC (µmhos/cm)	TDS (mg/L)
Value	12.30	0.00	12.30	86.20	82.60	40.40	522.00	0.00	129.00	65.60	1,310.00	700
DL	1.00	0.50	1.00	1.00	1.00	1.00	1.00	2.00	1.00	1.00	100.00	1

D. SOIL ANALYSES

Field 1

T3349020A

Sample and source description: T3349020A

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3349020B

Sample and source description: T3349020B

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field 2

T3349020C

Sample and source description: T3349020C

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			57.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field 2

T3349020D

Sample and source description: T3349020D

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field 3

T3349020D

Sample and source description: T3349020D

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			42.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3349020E

Sample and source description: T3349020E

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			37.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field 4

Field 4

T3349020F

Sample and source description: T3349020F

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			52.00		0.00	0.00	
DL			1.00		1.00	1.00	

T3349020G

Sample and source description: T3349020G

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			52.00		0.00	0.00	
DL			1.00		1.00	1.00	

Field 5

T3349020G

Sample and source description: T3349020G

Sample date: 12/15/2023 Source of analysis: Lab analysis

	Nitrate-N (mg/kg)	Total P (mg/kg)	Soluble P (mg/kg)	K (mg/kg)	EC (µmhos/cm)	Organic matter (%)	Total salt (mg/kg)
Value			52.00		0.00	0.00	
DL			1.00		1.00	1.00	

E. PLANT TISSUE ANALYSES

Field 3 - 10/27/2022: Oats, silage-soft dough

Field 3 - 10/27/2022: Oats, silage-soft dough

T3130026C

Sample and source description: T3130026C

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 75.2 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	12,000.00	2,380.00	18,400.00		12.90
DL	1.00	1.00	1.00		1.00

T3130026D

Sample and source description: T3130026D

Sample date: 05/10/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 73.8 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,600.00	2,360.00	18,600.00		20.60
DL	1.00	1.00	1.00		1.00

Field 5 - 05/15/2023: Corn, silage

T3265028A

Sample and source description: T3265028A

Sample date: 09/22/2023 Source of analysis: Lab analysis Method of reporting: Dry-weight

Moisture: 71.0 %

	Total N (mg/kg)	Total P (mg/kg)	Total K (mg/kg)	Total salt (mg/kg)	TFS (%)
Value	11,500.00	3,320.00	16,100.00		6.70
DL	1.00	1.00	1.00		1.00

F. SUBSURFACE (TILE) DRAINAGE ANALYSES

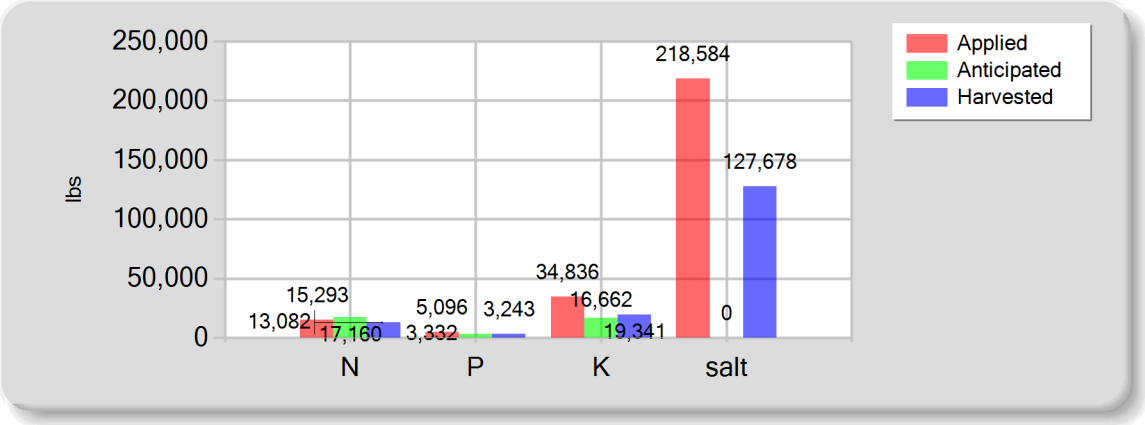
No subsurface (tile) drainage analyses entered.

NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

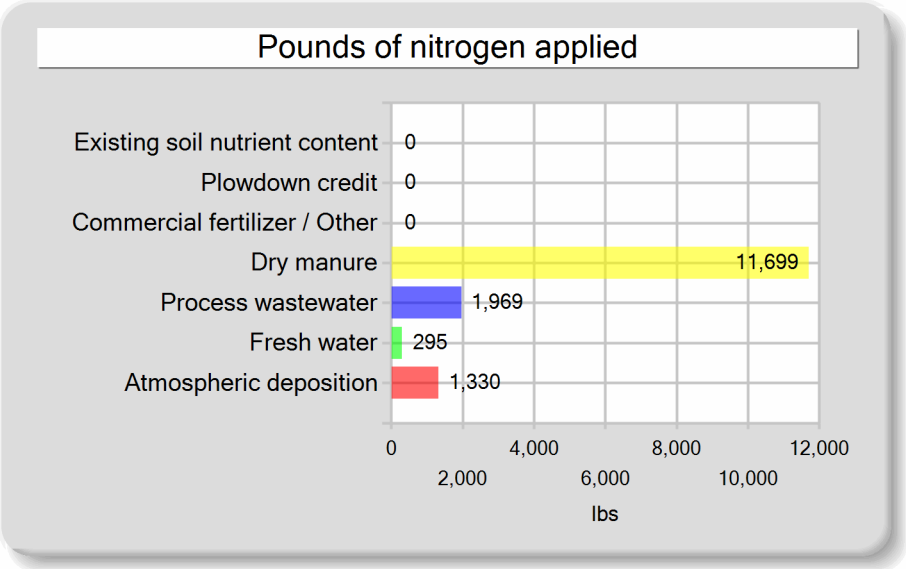
A. SUMMARY OF NUTRIENT APPLICATIONS, POTENTIAL REMOVAL, AND BALANCE

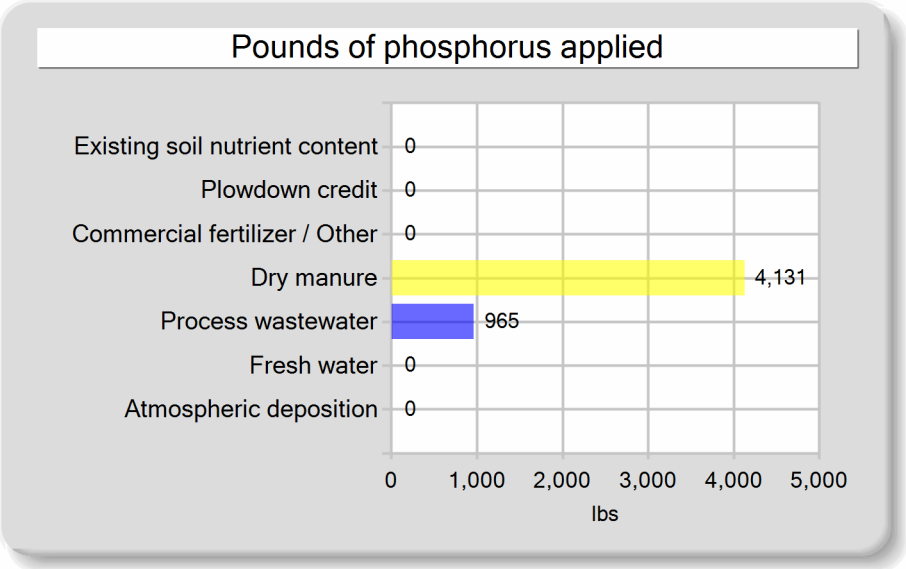
	Total N (lbs)	Total P (lbs)	Total K (lbs)	Total salt (lbs)
Existing soil nutrient content	0.00	0.00	0.00	0.00
Plowdown credit	0.00	0.00	0.00	0.00
Commercial fertilizer / Other	0.00	0.00	0.00	0.00
Dry manure	11,699.17	4,131.07	26,121.43	178,112.88
Process wastewater	1,968.78	964.52	8,714.68	40,471.58
Fresh water	295.00	0.00	0.00	0.00
Atmospheric deposition	1,330.00	0.00	0.00	0.00
Total nutrients applied	15,292.95	5,095.60	34,836.12	218,584.46
Anticipated crop nutrient removal	17,160.00	3,331.50	16,662.00	0.00
Actual crop nutrient removal	13,082.46	3,243.25	19,341.33	127,678.22
Nutrient balance	2,210.48	1,852.35	15,494.79	90,906.24
Applied to removed ratio	1.17	1.57	1.80	1.71

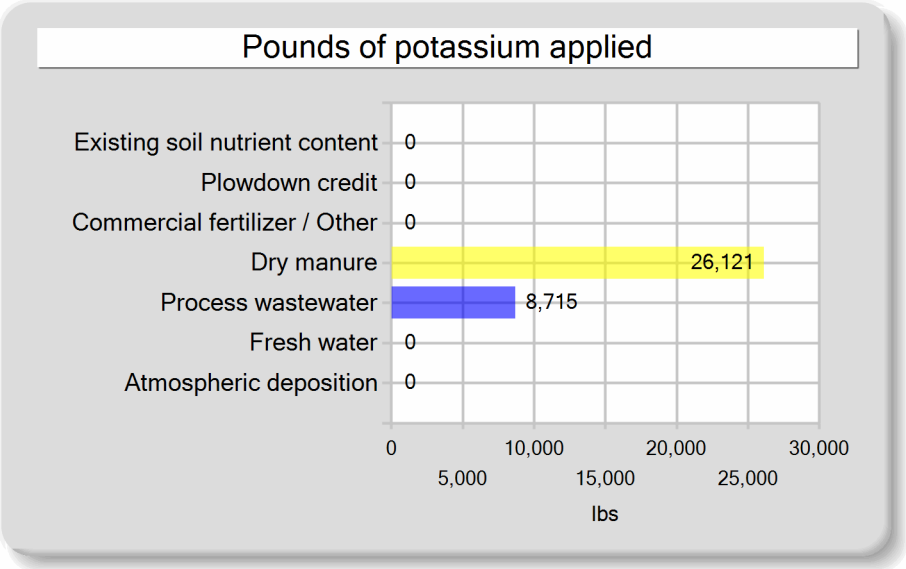
B. POUNDS OF NUTRIENT APPLIED VS. CROP REMOVAL

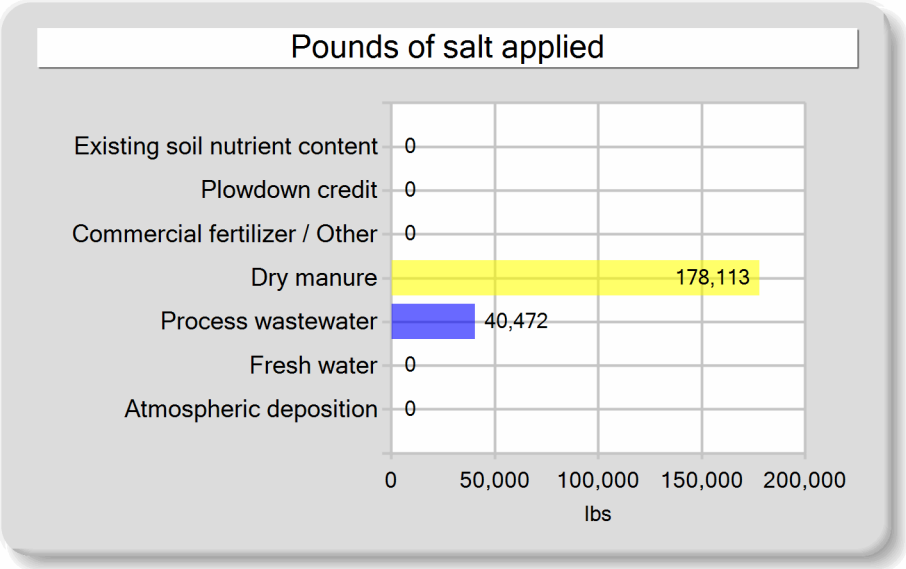


C. POUNDS OF NUTRIENT APPLIED BY MATERIAL TYPE









EXCEPTION REPORTING

A. MANURE, PROCESS WASTEWATER, AND OTHER DAIRY WASTE DISCHARGES

The following is a summary of all manure and process wastewater discharges from the production area to surface water or to land areas (land application areas or otherwise) when not in accordance with the facility's Nutrient Management Plan.

No manure or process wastewater discharges occurred during the reporting period.

B. STORM WATER DISCHARGES

The following is a summary of all storm water discharges from the production area to surface water during the reporting period when not in accordance with the facility 's Nutrient Management Plan.

No stormwater discharges occurred during the reporting period.

C. LAND APPLICATION AREA TO SURFACE WATER DISCHARGES

The following is a summary of all discharges from the land application area to surface water that have occurred during the reporting period when not in accordance with the facility's Nutrient Management Plan.

No land application area to surface water discharges occurred during the reporting period.

NUTRIENT MANAGEMENT PLAN AND EXPORT AGREEMENT STATEMENTS

A. NUTRIENT MANAGEMENT PLAN STATEMENTS

Was the facility's NMP updated in the reporting period?	Yes
Was the facility's NMP developed by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes
Was the facility's NMP approved by a certified nutrient management planner (specialist) as specified in Attachment C of the General Order?	Yes

B. EXPORT AGREEMENT STATEMENT

Are there any written agreements with third parties to receive manure or process wastewater that are new or were revised within the reporting period?	No

ADDITIONAL NOTES

A. NOTES

No notes entered for this annual report.

CERTIFICATION

A. OWNER AND/OR OPERATOR CERTIFICATION

I certify under penalty of law that I have personally examined and am familiar with the information submitted in this document and all attachments and that, based on my inquiry of those individuals immediately responsible for obtaining the information, I believe that the information is true, accurate, and complete. I am aware that there are significant penalties for submitting false information, including the possibility of fine and imprisonment.

George M. Simoes	George M Simoes
SIGNATURE OF OWNER OF FACILITY	SIGNATURE OF OPERATOR OF FACILITY
Simoes George	SAME AS OWNER
PRINT OR TYPE NAME	PRINT OR TYPE NAME
6-26-24	6-26-24
DATE	DATE

ATTACHMENTS

A. REQUIRED ATTACHMENTS

The following lists the required documents that should be attached to the Annual Report when submitted .

Annual Dairy Facility Assessment
Provide an Annual Dairy Facility Assessment (an update to the Preliminary Dairy Facility Assessment in Attachment A) for each reporting period. On the PDFA Final page, click on the ADFA Report button to generate an ADFA report after updating information as needed .

Manure/Process Wastewater Tracking Manifests
Provide copies of all manure/process wastewater tracking manifests for the reporting period, signed by both the owner/operator and the hauler.

Corrective Actions Documents
Provide records documenting any corrective actions taken to correct deficiencies noted as a result of the inspections required in the Monitoring Requirements of the General Order. Deficiencies not corrected in 30 days must be accompanied by an explanation of the factors preventing immediate correction.

Groundwater Monitoring
Dischargers that monitor supply wells or subsurface (tile) drainage systems, or that have monitoring well systems must submit monitoring results as directed in the General Order, Groundwater Reporting Section starting on page MRP-13.

Storm Water Monitoring
Dischargers that are required to monitor storm water more frequently than required in the General Order must submit monitoring results as directed in the General Order, Storm Water Reporting Section on page MRP-14.

DENELE ANALYTICAL, INC.

865 S. KILROY
TURLOCK, CA 95380
PH# 209-634-9055 FAX# 209-634-9057

DAIRY COMPLIANCE

Client: G. SIMOES DAIRY
Mailing address: 9792 DIABLO VISTA AVE.
(not facility address) GALT, CA 95632
Phone#: (209)744-9097

Authorized Copy Released To:
JCRK, INC.
(209)327-0992
Sampled By: George Simoes
Project ID:

ANALYSIS TO BE COMPLETED:

IRRIGATION/GROUND WATER
H1 EC*, NO3-N, NH4-N*
H6 EC, TN, TDS
H7 Combination of H1 & H6
H14 EC, NO3-N, NH4-N, Ca, Mg, Na, Bicarb Carb, SO4, Cl, TDS

PROCESS WASTE WATER
H3 EC*, NH4-N, TKN, TP, K, TDS
H4 EC*, NO3-N, NH4-N, TKN, TP, K, TDS
H5 Ca, Mg, Na, Bicarb, Carb, SO4, Cl
H8 Combination of H3 &H5
H9 Combination of H4 &H5

TILE DRAIN
H2 EC*, NO3-N, TP, NH4-N*, TDS

** Required Field Measurement*
___ (Check here if performed)

DISCHARGE WATER
E1 EC*, Temp*, pH*, NO3-N, NH4-N, TKN, TP, K, TDS, Dissolved Oxygen*, BOD 5, TSS, Total & Fecal Coliforms
E2 EC,* Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Dissolved Oxygen*, Total & Fecal Coliforms
E3 EC*, Temp*, pH*, NO3-N, NH4-N*, TP, Total & Fecal Coliforms
E4 EC*, Temp*, pH*, NO3-N, NH4-N*, Turbidity, TP, Total & Fecal Coliform

MANURE
M1 % Moisture
M3 TN TP, K, % Moisture
M4 Ca, Mg, Na, Bicarb, Carb, SO4, Cl, Ash
M7 Combination of M3 & M4

PLANT TISSUE
F18 % Moisture, Ash, TN, P, K, DM and as received
F18+ % Moisture, Ash, TN, P, K, DM and as received (Package also includes NIR DCAD)
F19+ % Moisture, TN (Package includes complete plant tissue analysis)

SOIL
K3 Soluble Phosphorus
K4 1ft NO3, Organic Matter
K5 Pre-Plant (2ft NO3), Fall (1-2ft NO3), Fall (2-3ft NO3)
K6 0-1ft EC, NO3, Soluble P, K, Organic Matter
K7 Combination of K3 & K4
K8 Combination of K3 & K5

	Date Collected	Time Collected	Sample ID	Description	Temp	Well Purge	H1	H3	M3	F18	OTHER
1	11/9/23	10:00a	Domestic Well #1	Dom Well	13.6°C	20 min					H14
2	✓	10:30a	Ag Well #1	Ag Well	12.0°	✓	X				
3	✓	11:00a	Ag Well #2	Ag Well	13.2°	✓	X				
4											
5											
6											
7											
8											
9											
10											

Analysis

CHAIN OF CUSTODY RECORDING

	Signature	Company	Received Date & Time	Relinquished Date & Time	(LABORATORY USE ONLY) Logged In By:	LABORATORY NUMBER
1st	Jacquie Capteim	JCRK		11/10/23 8:00a		
2nd		DENELE ANALYTICAL, INC.	11-10-23 1100			T3314016

Required to be submitted to any agency? (YES / NO)

Denele Analytical, Inc.

865 South Kilroy Avenue Turlock, CA 95380
Phone (209) 634-9055 - Fax (209) 634-9057

www.denelelabs.com

40 N. East Street, Woodland, CA 95776
Phone (530) 666-9056 - Fax (530) 666-9058

Compliance Analysis Report

Customer
G. Simoes Dairy Inc.
9792 Diablo Vista Ave.
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/13/23

Irrigation /Ground Water (H14)
SOURCE: Well Water

NOTES: Temperature was above qualifying at 13.6c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
DW #1	T3314016A	Electrical Conductivity (EC)	EPA 120.1 and	1.31	mmhos/cm
		Bicarbonate Alkalinity (ALK-B)	SM 2320B-2011	522	mg/L
		Carbonate Alkalinity (Carb)	SM 2320B-2011	< 2.00	mg/L
		Sulfate (SO4)	EPA 375.2	129	mg/L
		Sodium (Na)	EPA 200.7	40.4	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Magnesium (Mg)	EPA 200.7	82.6	mg/L
		Nitrate Nitrogen (NO3-N)	EPA 353.2	12.3	mg/L
		Total Dissolved Solids (TDS)	SM 2540 C-1997 and	700	mg/L
		Calcium (Ca)	EPA 200.7	86.2	mg/L
		Chloride (Cl)	SM 4500-Cl D-2011	65.6	mg/L
		Soluble Salts (SALT-SOL)	Calculation	836	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	Sample mg/L	Dup mg/L	RPD %
Ammonia Nitrogen (NH3-N)	94.4			0.000 (T3314013A)	0.000 (null)	
Calcium (Ca)	105	94.5 (T3314002A)	94.4 (T3314002A)			0.102
Chloride (Cl)	101	91.1 (T3314001A)	90.2 (T3314001A)			0.927
Magnesium (Mg)	94.1	103 (T3314002A)	102 (T3314002A)			1.14
Nitrate Nitrogen (NO3-N)	100	106 (T3314001A)	103 (T3314001A)			3.44
Sodium (Na)	93.2	110 (T3314002A)	107 (T3314002A)			2.17
Sulfate (SO4)	108	126 (T3314001A)	126 (T3314001A)			0.615
Total Dissolved Solids (TDS)	102			50.0 (T3313017B)	50.0 (Dup)	0.000

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.

ELAP Certificate No.: 2714

Denele Analytical, Inc.

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Compliance Analysis Report

Customer
G. Simoes Dairy Inc.
9792 Diablo Vista Ave.
Galt, CA 95632

Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 12.0c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well #1	T3314016B	Electrical Conductivity (EC)	EPA 120.1 and	0.616	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	2.52	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	394	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

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Compliance Analysis Report

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Grower

Consultant:
Jacquie Captein
P.O. Box 386
Herald, CA 95638

PURCHASE ORDER:
RECEIVED DATE: 11/10/23 1:00 PM
SUBMITTED BY: Jacquie
ANALYZED DATE: 11/14/23

Irrigation/ Ground Water (H1)
SOURCE: Irrigation Water

NOTES: Temperature was above qualifying at 13.2c

Sample ID	Lab ID	Analyte	Method Ref	Result	Units
Ag Well #2	T3314016C	Electrical Conductivity (EC)	EPA 120.1 and	0.510	mmhos/cm
		Nitrate Nitrogen (NO3-N)	EPA 353.2	1.23	mg/L
		Ammonia Nitrogen (NH3-N)	SM 4500-NH3 D,C	< 0.500	mg/L
		Soluble Salts (SALT-SOL)	Calculation	326	mg/L

Laboratory Quality Control	LCS %	MS %	MSD %	RPD %
Ammonia Nitrogen (NH3-N)	92.6			
Electrical Conductivity (EC)	93.2			1.39
Nitrate Nitrogen (NO3-N)	102	100 (T3314015D)	100.0 (T3314015D)	0.459

The warranty of Denele Analytical is limited to the accuracy of the analyses of the samples as received. Denele Analytical assumes no responsibility for which the customer uses our test results, nor liability for any other warranties, express or implied. These terms and conditions shall supercede any conflicting terms and conditions submitted on customer purchase orders or other forms submitted for work.